

Extraction and isolation of pectin rich in homogalacturonan domains from two cultivars of hawthorn berry (*Crataegus pinnatifida*)

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Abstract

Hawthorn berries present high pectin content with good ability to form and stabilize oil-in-water emulsions. However, successful applications are determinant of high yield and purity as well as good surfactant properties based on their molecular structure. In this work, pectins from one of the most common *Crataegus pinnatifida* cultivars (Dajinxing, DA), and from a less known cultivar gaining popularity (Mianqiu, MI), were extracted and isolated. They were comparatively investigated with two commercial pectins from citrus peel [$> 67.9\%$ degree of methyl-esterification (DM) and $> 78\%$ galacturonic acid (GalA)] and analyzed for their yield, molecular weight (HPSEC-MALS-RI), monomer composition and glycosyl linkage analysis (GC-MS), DM (titration, FTIR and $^1\text{H-NMR}$), and emulsifying performance. MI showed a 2-fold greater pectin yield (in terms of GalA) than DA (61.2% vs. 38.3%). Thus, hot acid extraction had to be followed by an enzymatic step to remove starch molecules and obtain purified DA and MI pectins ($>86\%$ GalA), which also resulted in pectin demethoxylation (DM $<38\%$). Remarkably, purified hawthorn pectins presented lower molecular weight than commercial citrus pectins (98-126 kDa vs 164-235 kDa) together with a higher proportion of homogalacturonan (HG) domain and a significantly greater emulsifying activity and stability. Specifically, MI showed a higher HG proportion and a less branched rhamnogalacturonan-I (RG-I) domain [Branching calculated as $(\text{Arabinose}+\text{Galactose})/\text{Rhamnose}=2.2$] than DA (Branching=3.1), with significantly shorter arabinan and galactan side chains (evidenced by more terminal arabinose and galactose). The yield and structure of MI pectin strongly indicates that MI berry is an excellent pectin source with distinguished functionality.

Keywords: hawthorn; pectin; emulsion, rhamnogalacturonan; homogalacturonan; branching degree

1 Introduction

Hawthorn (*Crataegus* spp.) is a large genus of thorny shrubs or small trees distributed in the Northern Hemisphere, mostly in China, Europe and North America. The production of the hawthorn fruit has grown significantly in recent years because of increased intake as fresh fruit or after being processed into juice and jams. Interestingly, the fruit of hawthorn has a higher pectin content than other cultivated fruits (Cuevas-Bernardino, Lobato-Calleros, Román-Guerrero, Alvarez-Ramirez, & Vernon-Carter, 2016; Li, Li, Wang, & Liu, 2008; Li, Hu, & Xu, 2015), and its pectin has been reported to exhibit six-fold higher viscosity (Li et al., 2008) and to be more efficient at forming and stabilizing oil-in-water emulsions than commercially available pectin from citrus peel (Cuevas-Bernardino et al. 2016). In addition, hawthorn pectic polysaccharides and oligosaccharides have also showed health benefits towards the cardiovascular system, including cardiovascular protection, endothelium-dependent vasorelaxation, improvement of coronary circulation, and hypolipidemic effects (Li, Huang, Dong, Zhu, & Li, 2017; Zhu et al., 2015; Zhu et al., 2017).

Pectin is a heteropolysaccharide consisting mainly of two major regions, namely, homogalacturonan (HG) and rhamnogalacturonan I (RG-I). Homogalacturonan (known as the smooth region) is the most abundant segment and it is composed of long chains of linearly α -1,4-linked D-GalAp residues (galacturonic acid, GalA) (Kpodo et al., 2018). Some of the carboxyl groups are methyl-esterified at C-6 position and it is possible that GalpA is also acetyl esterified at O-2 and/or O-3 positions. The RG-I (known as the hairy region) is composed of the repeating GalA and rhamnose (Rha) disaccharide; $[\alpha$ -1,2-linked-L-Rhap- α -1,4-D-GalAp]_n where n can be greater than 100 (Chan, Choo, Young, & Loh, 2017). Rhamnose units on the RG-I segment contain polymeric side-chains predominantly composed of the neutral sugars arabinose, galactose or both. Other

moieties may be observed depending on the botanical origin and method of extraction, such as rhamnogalacturonan II (RG-II), arabinogalactan, arabinan, apiogalacturonan, xylogalacturonan or proteins attached to side chains of RG-I regions.

Most polysaccharides do not perform well as emulsifying agents because of their reduced hydrophobicity (Kpodo et al., 2018). However, pectins extracted from sugar beet (Schmidt, Schmidt, Kurz, Endress, & Schuchmann, 2015), pumpkin (Cui, & Chang, 2014), okra (Alba, Ritzoulis, Georgiadis, & Kontogiorgos, 2013; Xu, Martinez, Yang, & Guo, 2020) and hawthorn (Cuevas-Bernardino et al., 2016) have been reported to form and stabilize emulsions. The emulsifying properties of sugar beet and okra pectins have been generally attributed to either high acetyl content, presence of covalently bound proteins, a lower proportion of RG-I, and RG-I with branches of intermediate length (Kpodo et al., 2018). On the other hand, the emulsifying properties of hawthorn pectins have been attributed to the adequate balance between high molecular weight and hydrophobic contributing moieties (high degree of methyl esterification, DM) in their molecular structure (Cuevas-Bernardino et al., 2016), the former also fostering their use as thickening agent. Thus, mechanistic understanding connecting detailed information about the molecular structure of hawthorn pectin with its functionality is still missing.

More than 1000 species of the genus *Crataegus* spp. have been described, among which mountain hawthorn, also known as Chinese hawthorn (*Crataegus pinnatifida*) is the most widely distributed in China (Guo & Jiao, 1995). In addition to the natural species, many varieties and cultivars are available as a result of long-term cultivation and plant breeding (Zhao & Tian, 1996). The isolation of pectin from a single hawthorn cultivar has been previously reported, showing GalA values from 68 to 90.2 % (Cuevas-Bernardino et al., 2016; Guo, Du, Jiang, Goff, & Cui, 2019; Jiang, Du, Zhang, & Li, 2018). The big differences in the purity (GalA content) could be attributed to the extraction procedure,

cultivar, or fruit maturity, which determines the presence of other co-extracted biopolymers, such as starch. Notably, starch has been reported to be present in hawthorn berries depending on the fruit maturity (Li et al., 2015), which could gelatinize, solubilize and be co-extracted during hot acid extraction of pectin. Therefore, small variations in the complex molecular structure of hawthorn pectin between cultivars and extraction procedures could significantly determine its ultimate functional properties, including their suitability as emulsifiers.

In this work, pectin extracted from one of the most common cultivars of *Crataegus pinnatifida* in China, namely Dajinxing (DA) cultivar, and from a less common cultivar that is gaining popularity for its larger fruits of softer texture, namely Mianqiu (MI), was extracted using 2 different procedures and comparatively investigated for their yield, hydrodynamics, fine molecular structure (including monomer and glycosyl linkage composition) and emulsifying performance. This work also aims at providing structural information of hawthorn pectins that can be used to select best hawthorn cultivars for optimal pectin yield and molecular structure. The emulsifying results are expected to serve as a proxy for future complete colloidal studies.

2. Materials and methods

2.1 Materials

Fresh hawthorn (*Crataegus pinnatifida*) berries from Dajinxing (DA) and Mianqiu (MI) cultivars were manually collected on the 138th day after full bloom in the central mountainous region of Shandong province (36°04' N-36°37' N ; 118°14' E-118°49' E) and supplied by Yijia Food Co. (Linqu, Shandong, China). Commercially available high methoxyl citrus pectins were gently provided by Ceamsa (Porriño, PO, Spain). Ceampectin ESS 4400 (CP1) and RS 4700 (CP2) presented 61 % and 71 % of degree of methyl esterification, respectively (data provided by the manufacturer).

Amyloglucosidase from *Aspergillus niger* (E-AMGDF, 10 mL, 3,300 U/mL) was obtained from Megazyme (Megazyme Ltd. Wicklow, Ireland). Type VI-B α -amylase from porcine pancreas (10 U/mg) was obtained from Sigma Aldrich (Sigma Chemical Co., St. Louis, Missouri). All other reagents were of analytical or HPLC grade.

2.2 Methods

2.2.1 Isolation of pectic polysaccharides

After removing the seeds, hawthorn berries were cut into slices of 1-2 mm thickness and dried at 40 °C until the moisture was below 10 %. Dried berries were ground into powder using an electric grinder (SinoNaichi Technology Co., LTD, Beijing, China) and sieved to pass a screen of 270 μ m to obtain hawthorn flours. Pectin content in hawthorn flours was estimated using the m-hydroxydiphenyl method for GalA content (Kintner & Buren, 1982). Furthermore, soluble dietary fiber precipitated with ethanol (SDFP) was measured according to AACC 32-50.01 method (AACC, 2015) using the Integrated Total Dietary Fiber assay from Megazyme.

Pectin was extracted from hawthorn flours following the method of Jiang et al. (2018), as shown in Figure 1. Hawthorn flour (10.0 g) was washed with 95 % ethanol (1:15 w/v) for 2 h ($\times$ 3 times) under continuous stirring to remove non-cell wall components. After centrifugation (4500 rpm, 10 min, 20 °C), the alcohol-insoluble residue (AIR) was washed with 200 mL acetone and centrifuged to remove any alcohol soluble low molecular weight components, such as phenolics and sugars, and minimize the activity of cell wall degrading enzymes. The AIR pellet was air-dried at ambient temperature overnight for pectin extraction. Since highest yields of pectic substances are generally obtained by hot acid extractions, which is also the most convenient approach for industrial extraction of pectin, AIR was mixed with distilled water (1:20 w/v), then the pH was adjusted to 2.0 using a citric acid solution (4 mol/L) and incubated at 90 °C for 2 h under continuous stir.

The extract solution was separated from the precipitate by centrifugation (4500 rpm, 10 min, 20 °C), and cooled down to room temperature. Afterwards, the supernatant was mixed with technical ethanol (95 %) until 70 % saturation. The mixture was kept at 4 °C overnight after stirring gently for 15 min. A pectin-rich residue (residue 1) was obtained by centrifugation as above (4500 rpm, 10 min, 20 °C). This pectin-rich residue was used to obtain both crude and purified pectins (Figure 1). The preliminary results from the neutral sugar composition (detailed in section 2.2.2) denoted the presence of a high amount of glucose, which highly suggests the presence of co-eluted starch in the crude pectin (Pettolino et al., 2012). This finding was supported by previous results in literature (Guo et al., 2019; Li et al., 2015) and set the reason for performing the purification protocol using starch hydrolytic enzymes. Thus, crude and purified pectin (after enzymatic starch removal) were extracted and comparatively investigated in this work. Pectin purification was performed as reported in Chigwedere et al. (2019) with slight modifications (Figure 1).

For the preparation of purified pectin, pectin-rich residue 1 was suspended in 50 mL of sodium acetate buffer (0.1 mol/L, pH = 4.3) and incubated with 0.5 mL of *Aspergillus niger* amyloglucosidase and 0.16 g of porcine pancreas Type VI-B α -amylase at 62.5 °C for 4 h. Samples were then cooled down in an ice bath and technical ethanol (95 %) was added to 70 % saturation followed by resting at 4 °C overnight (pectin-rich residue 2). Residue 1 (for the preparation of crude pectin) or residue 2 (for the preparation of purified pectin after starch removal) were washed twice with 50 mL of 95 % ethanol and acetone (as described before), and dissolved into deionized water and dialyzed for 72 h using a dialysis bag of 3.5 kDa molecular weight cutoff. Crude and purified (+E) pectins from DA and MI hawthorn cultivars were finally obtained by freeze-drying.

2.2.2 Chemical characterization

Moisture was determined according to the AACC approved method 44-15.02 (AACC, 2015). GalA content was determined using the m-hydroxydiphenyl method described by Kintner and Buren (1982) using D-galacturonic acid as standard. Crude protein was measured using Dumas combustion method (AACC 46-30.01) with the factor of 6.25 to convert the measured nitrogen into protein (Cuevas-Bernardino et al., 2016). Results of GalA and protein content were expressed as weight percentage in dry basis (d.b.).

The degree of methyl esterification (DM) was determined by the titrimetric method (Kazemi, Khodaiyan, Labbafi, Hosseini, & Hojjati, 2019). 100 mg of pectin was mixed with 2 mL ethanol and 20 mL CO₂-free water. After complete dissolution of the pectins, three drops of phenolphthalein reagent were added and the solution was titrated with NaOH (0.1 mol/L) until the appearance of pink color (first volume, V_1). Afterwards, 10 mL of NaOH (0.1 mol/L) was added to the solution and stirred for 20 min. Then, 10 mL of HCl (0.1 mol/L) was added and shaken until complete disappearance of the pink color. Finally, the solution was titrated with NaOH (0.1 mol/L) until a faint pink color was obtained that persisted after vigorous shaking (second volume, V_2). The DM of pectin was calculated as follows:

$$\text{DM (\%)} = V_2 \times 100 / (V_2 + V_1)$$

Neutral sugars composition was analyzed following the method of Pettolino, Walsh, Fincher and Bacic (2012). Pectin-rich samples were hydrolyzed in 2.5 mol/L trifluoroacetic acid for 2 h at 121 °C. After acid hydrolysis, acid was removed by evaporating samples to dryness with a stream of nitrogen and subsequent dissolution of the dry material in methanol, followed by another drying step. Drying with methanol was repeated two more times to ensure complete elimination of the acid. Then, samples were subjected to reduction with a solution of 1 mol/L sodium borodeuteride in 2 mol/L ammonium hydroxide. Reduced monosaccharides were quantified as alditol acetates

derivatives in acetone using a GC/MS system (7890A and 5975C inert MSD with a Triple-Axis detector, Agilent Technologies, Inc., Santa Clara, CA) equipped with a SP-2330 capillary column (30 m x 0.25 mm ID x 0.20 μ m, SUPELCO, Bellefonte, PA). 1 μ L of samples were injected in split mode 2:1 and eluted using helium as carrier at a flow rate of 1 mL/min by applying the following oven temperature settings: the temperature was initially at 180 °C, then increased at 15 °C/min to 248 °C and hold for 7 min. A solvent delay of 4 min was used. The injector, ion source and quadrupole mass analyzer were kept at 250 °C, 230 °C and 150 °C, respectively. Calibration curves were performed with standard solutions of rhamnose (Rha), fucose (Fuc), arabinose (Ara), xylose (Xyl), mannose (Man), galactose (Gal) and glucose (Glc). The monosaccharide content was expressed as mol % of the total monosaccharide. Myo-inositol was used as internal standard in all derivatizations. All measurements were performed in triplicate.

2.2.3 Glycosyl linkage composition

Linkage analysis began with solubilization of the samples in DMSO for 48 h prior the methylation of all free hydroxyl groups in the intact polysaccharide following the procedure described in Pettolino et al. (2012). Partially methylated monosaccharides were analyzed in the form of partially methylated alditol acetates, performing reduction and acetylation as aforementioned for monomer composition. Samples were quantified by GC/MS on a SP-2330 capillary column using the same chromatographic conditions as those for neutral monomer composition, but with a different temperature program for better resolution of the peaks: initial temperature of 140 °C for 2 min, and then 10 °C/min to 240 °C for 9 min. Triplicate samples were used for monosaccharide composition and linkage analyses. Identification of glycosyl linkages was performed based on a library of standards and quantification was based on the amount (mol %) of each individual monomer. For glycosyl linkage composition, *f* and *p* after each individual monomer (Rha,

Fuc, Ara, Xyl, Man, Gal and Glc), indicate the presence of the neutral sugar in the furanose and pyranose form, respectively. The numbers before each monomer (i.e., 1,4-Glc) indicate the type of linkage. T- before a monomer indicates the presence of a terminal linkage, meaning that this glycosyl residue is only linked at C-1.

2.2.4 Spectroscopic analyses

Fourier transform infrared (FT-IR) spectra of hawthorn and citrus samples were obtained between 600 and 4000 cm^{-1} in attenuated total reflection (ATR) mode at a resolution of 4 cm^{-1} using 32 co-added scans (DTGS detector, Vertex 70, Bruker, USA). The measurement was performed on a Ge ATR crystal at a 45-degree incident angle. In the FT-IR spectrum bands at $\sim 1730 \text{ cm}^{-1}$ and 1630 cm^{-1} are derived from methyl-esterified and free carboxyl groups, respectively. Thus, the relative proportion of the 1730 peak area (A_{1730}) to the total area of peaks A_{1730} and A_{1630} [$A_{1730}/(A_{1730} + A_{1630})$] was used to estimate the degree of methyl esterification (DM). Spectral smoothing and background removal was applied using instrument software for peak area analysis.

^1H -NMR analysis was conducted using a Bruker Avance III NMR (Bruker Co., Switzerland) at 400 MHz. Prior to analysis, 15 mg of samples were dispersed overnight in 1.5 mL of D_2O (10 % w/v) containing 3-(Trimethylsilyl)-1-propanesulfonic acid- d_6 sodium salt (DSS) as internal standard in a thermomixer set at 70 $^\circ\text{C}$ and 350 rpm. ^1H -NMR spectra were recorded with the zg30 pulse program of the NMR at 70 $^\circ\text{C}$. The acquisition parameters were the following: spectral width 5597.015 Hz, relaxation delay 10 s, number of scans 16 and acquisition time 3.9998 s. TopSpin 4.0.8 software (Bruker Co., Switzerland) was used for analysis of recorded spectra.

2.2.5 Molecular weight determination

The molecular weight (M_w) of pectin molecules was measured using size exclusion chromatography (SEC) coupled to a multi-angle light scattering (MALS) detector (Dawn

Heleos, Wyatt Technology, Santa Barbara, CA, USA) containing a K5-cell following the method of Kpodo et al. (2017), with slight modifications. The system also included a refractive index (RI) detector (OptilabrEX). Diluted (3 mg/mL) dispersions of the pectin samples were prepared in 50 mmol/L sodium nitrate mixed for 2 h, and allowed to hydrate overnight at 40 °C. Subsequently, samples were filtered through a 5 µm nylon membrane filters and transferred into the vials. 100 µL of sample were injected into the HPSEC-MALS-RI system equipped with PL aquagel-OH 60 8 µm, PL aquagel-OH 50 8 µm and PL aquagel-OH 40 8 µm columns (Agilent Technologies, Waldbronn, Germany) connected in series and running at a temperature of 35 °C. Samples were eluted with degassed and filtered sodium nitrate (50 mmol/L, pH 5.8) as a mobile phase at a flow rate of 0.7 mL/min. Data was analyzed based on the light scattering theory using ASTRA software (version 5.3.4.14, Wyatt Technology Corporation, Goleta, CA, USA) following the Berry second order plot procedure. The specific refractive index increment (dn/dc) was assumed as 0.146 mL/g, value reported for pectin dissolved in sodium nitrate.

2.2.6 Emulsifying properties

Using the methods of Yasumatsu et al. (1972) and Jeddou et al. (2016) with the modifications reported in Xu et al. (2020), 10 mL of 0.5 % (w/v) pectin solutions and 10 mL of refined sunflower oil (Mazola, USA) were mixed uniformly in 50 mL conical centrifuge tubes. The mixture was emulsified using a homogenizer (IKA-T18, Werke GmbH & CO., Germany) at 10,000 rpm for 1 min. The emulsions were centrifuged at $1300 \times g$ for 5 min. Emulsifying ability (EA) was calculated as follows: $EA (\%) = (\text{the height of emulsified layer} / \text{total height of whole layer in centrifugal tube}) \times 100$.

To measure the emulsion stability (ES), the emulsion prepared by the procedure for EA determination was heated at 80 °C for 30 min, cooled to ambient temperature and

centrifuged at $1300 \times g$ for 5 min. ES was calculated as follows: $ES (\%) = (\text{the height of remaining emulsified layer} / \text{total height of whole layer in centrifugal tube}) \times 100$

2.2.7 Statistical analysis

Statistical analysis was conducted using Statgraphics Centurion XVI software (Statpoint Technologies, Inc., The Plains, VA, USA). Significant differences ($p < 0.05$) between hawthorn parameters were studied by analysis of variance (ANOVA) and results were expressed as mean values $\pm$ standard deviation. Fisher's Least Significant Difference (LSD) was used to describe means with 95 % confidence intervals.

3 Results and discussion

3.1 Compositional characterization of hawthorn and commercial citrus pectins

The pectin content (expressed as weight percentage of GalA) in the dried hawthorn berries was $7.8 \pm 0.5 \%$ and $14.2 \pm 1.4 \%$ for Dajinxing (DA) and Mianqiu (MI) cultivars, respectively. These values were similar to the content in soluble dietary fiber precipitate (SDFP), where DA and MI displayed $9.4 \pm 0.1 \%$ and $14.5 \pm 1.9 \%$, respectively. This would indicate that MI cultivar possesses berries with almost two-fold higher pectin content. In fact, although the yield of crude pectin material after extraction using hot buffer at pH 2.0 in citric acid was almost the same for both cultivars (10.79 - 10.65 % of freeze-dried pectin recovered, that is, 10.79 - 10.65 g of pectic material recovered for 100 g of hawthorn berry flour), MI crude pectin exhibited an almost two-fold higher purity, with GalA contents of 61.2 and 38.3 % for MI and DA, respectively (Table 1). Thus, the combined results for the total extraction yield and the GalA content of the final pectic material recovered showed a higher yield of purified pectin extracted in MI cultivar (Alba, Laws, & Kontogiorgos, 2015; Pereira et al., 2016). This occurrence will also be supported by the presence of co-extracted glucose-based polysaccharides, such as starch (Table 2).

Table 1 shows the chemical composition of crude and purified hawthorn pectins, namely without (DA and MI) and with enzymatic treatment for starch removal (DA +E and MI +E) after crude pectin extraction. For comparative purposes, two high methoxyl commercial citrus pectins, CP1 and CP2, were also included in this work. The GalA content of the crude hawthorn pectins varied from 38.3 to 61.2 % (as explained before), indicating that high molecular weight soluble carbohydrates were co-extracted in the crude pectins of both cultivar, and especially in that from DA. The difference between the pectin content (7.8 %) and SDFP (9.4 %) in DA flour suggests the presence of around 1.6 % of other co-extracted soluble non-starch polysaccharides, that may include hemicelluloses (i.e., xyloglucans), in DA crude pectin, whereas these would not be present in appreciable quantities in MI crude pectin. In addition, results from monomer composition (glucose content) and glycosidic linkage analyses (glucose monomers mainly as α -1,4-linked-D-Glucp, 1,4-Glucp in Table 2) pointed to the presence of significant amounts of co-extracted starch in both DA and MI crude pectins. As starch is present in hawthorn berries (Li et al., 2015), it could gelatinize, solubilize and be co-extracted during hot acid extraction of pectin. Thus, after the enzymatic hydrolysis step to remove starch in crude pectins, the GalA content increased from 38.3 % and 61.2 % to 83.2 and 85.3 % for DA+E and MI+E, respectively, obtaining similar purity to CP1 (77.4 %) and CP2 (86.4 %). Similar purities were obtained when extracting pectin from hawthorn wine pomace (Jiang et al., 2018). The small differences in purities between our hawthorn pectins and those from other studies (Guo et al., 2019; Jiang et al., 2018) can be due to differences in the extraction procedure as well as in the maturity state of the hawthorn berries, leading to compositional differences in the carbohydrates fraction present in the fruit (Li et al., 2015). Purified pectins exhibited a protein content always lower than 4.4 % (dry basis), which agrees with other purified citrus and hawthorn pectins

reported elsewhere (Cuevas-Bernardino et al., 2016). Among hawthorn cultivars, DA always presented higher protein content than MI, both crude and purified, suggesting a higher protein content in the initial berry flour. The presence of proteinaceous components could also originate from arabinogalactan-protein complexes (Immerzeel, Eppink, de Vries, Schols, & Voragen, 2006).

The neutral sugar composition and their linkage pattern (Table 2) provided a detailed representation of the pectin molecular structure and served for the identification of other non-pectin polysaccharides in crude and purified pectins. Higher amounts of neutral sugars were reported for DA and MI (65.8 and 42.5 % weight, respectively) followed by CP1 and CP2 (13.8 and 12.8 % weight, respectively), and, purified DA+E and MI+E hawthorn pectins (9.6 and 8.1 % weight, respectively). Neutral sugar composition analysis (% mol) showed the presence of rhamnose (Rha), arabinose (Ara), galactose (Gal), glucose (Glc), xylose (Xyl), mannose (Man) and fucose (Fuc) in all studied samples (Table 2 and Figure 2). In Figure 2A, it was clearly observed that the main neutral sugar present in crude pectins DA and MI was glucose, with 58.90 and 35.27 %, respectively. 84 % of this glucose consisted of 1,4-Glcp, and 7 % of 1,4,6-Glcp (Table 2), confirming that this glucose belonged to linear and branched starch molecules, respectively (Bertoft, 2017; Pettolino et al., 2012). Furthermore, after pectin purification with starch hydrolytic enzymes (DA+E and MI+E), this glucose was reduced down to 1.43-1.28 %, clearly showing that significant amounts of starch were co-extracted in crude materials and eliminated after purification (< 1.5 %).

In Figure 2B, neutral sugars of purified hawthorn pectins were compared with commercial citrus pectins. The main detected sugars in DA+E and MI+E were galactose (especially in DA+E), arabinose and rhamnose, which were likely derived from the rhamnogalacturonan I (RG-I) fraction in pectin molecules. These monomers were also

the most abundant ones in CP1 and CP2, with CP1 being especially rich in arabinose and CP2 in galactose. Very low levels of glucose and xylose were detected in the purified citrus and hawthorn pectins, suggesting a low but detectable presence of rhamnogalacturonan II (RG-II) or xylogalacturonan regions (Alba et al., 2015). The low amounts of fucose and mannose also suggests a small presence of xyloglucans and mannans, respectively. It is noteworthy that xylans and mannans have been detected in other berries, such as blackcurrants (Hilz, de Jong, Kabel, Schols, & Voragen, 2006) and grapes (Minjares-Fuentes et al., 2016). However, data from glycosyl-linkage composition showed 100 % of the xylose and mannan residues as Terminal (T-Xyl_p and T-Man_p, respectively), which are typically found in xyloglucans or xylogalacturonans and RG regions (Pettolino et al., 2012; Shiga & Lajolo, 2006). Therefore, results discarded the presence of xylans and mannans in hawthorn crude and pure pectin fractions. Of note that xylose, mannose and glucose were more abundant in hawthorn pectin than in citrus counterparts.

Glycosyl-linkage composition of crude and purified hawthorn pectins revealed the same type of linkages after both extraction procedures, with more than 20 types of linkages pattern. Therefore, glycosyl-linkage composition of monomers from purified hawthorn pectins will be discussed as they also offer more reliable results due to the higher purity. RG-I consists of repeating units of the disaccharide α -1,2-linked-L-Rhap- α -1,4-D-GalAp. In most cases, 20–80 % of rhamnose residues in this domain are substituted at the C-4 position with neutral sugar side chains (Mohnen, 2008) mainly formed by α -Araf (arabinans), β -Gal_p residues (galactans) and/or a mix of the two (arabinogalactans) (Chan et al., 2017; Guo et al., 2019). In this study, the presence of 1,2-Rhap and 1,2,4-Rhap indicated the existence of RG-I region. In addition, 1,2,4-Rhap represented 44 and 35 % of the rhamnose residues in DA+E and MI+E, respectively, indicating 44 (DA+E) or 35

% (MI+E) of the rhamnose residues were branched with arabinan/galactan/arabinogalactan side chains at the O-4 position. Arabinose residues were mainly found to be as 1,5-linked *Araf* (arabinofuranose), followed by T-*Araf*, whereas galactose was found as 1,4-linked *Galp* (galactopyranose), and T-*Galp*. The linkage pattern of arabinose and galactose was in agreement with previous data from Guo et al. (2019) investigating hawthorn pectin (68 % GalA). Smaller amounts of 1,3-*Galp* and 1,6-*Galp* were also observed, which were previously identified as part of arabinogalactan side chains (Mohnen, 2008; Shiga & Lajolo, 2006). 1,3-*Araf* was also found in minor amounts, suggesting a minor amount of branched types of linkages in arabinans or arabinogalactans. Similarly, no branched types of linkages were detected concerning galactose, which might be related to their lower amount that probably was under the limit of detection. For both arabinan and galactan side chains, terminal arabinose and galactose were more abundant in MI+E than in DA+E, probably indicating RG-I segments with shorter side chains in MI cultivar. Arabinose in the form of T-*Arap* (arabinopyranose) and 1,2-*Arap* was also observed, the former also detected by Guo et al. (2019) in hawthorn pectin and the latter being reported in RG-II regions (Mohnen, 2008). The small presence of T-*Glc**p*, 4-*Glc**p*, 4,6-*Glc**p*, T-*Galp*, 2-*Galp*, T-*Fuc**p* and T-*Xylp* linkages also suggests the presence of some xyloglucans and/or xylogalacturonan regions (Shiga & Lajolo, 2006). It is noteworthy that small proportions of other sugars such as glucose, mannose, fucose, xylose, and glucuronic acid are found covalently linked to the backbones along the RG-I region as side chains (Chan et al., 2017). Thus, the presence of mannose residues as T-Man suggests they may be part of galactan side chains rather than part of mannans (Shiga & Lajolo, 2006).

The molar ratios of sugars in purified pectins were used to further investigate the structure of the purified pectins at the molecular level (Table 3). The HG molar percentage of the

hawthorn pectins ranged from 86.3 to 88.0 %, whereas that of RG-I varied between 8.7 to 10 %. Slightly lower HG (79 – 83 %) and significantly higher RG-I (14.7 - 18.7 %) values were observed in citrus pectins, especially in CP1, showing that extracted hawthorn pectin was essentially a linear pectin due to the predominance of the HG region. In fact, hawthorn pectin showed a higher proportion of HG than most reported pectins, including okra, with $HG \leq 45\%$ (Alba et al., 2015; Kpodo et al., 2018), blackcurrant (66-77 %, Alba et al., 2018) and apple and citrus pectins (~70.8-77.8 %, Leroux, Langendorff, Schick, Vaishnav, & Mazoyer, 2003). In this sense, Guo et al. (2019) also indicated that HG was the main region in hawthorn pectin. Interestingly, Jiang et al. (2018) also found high proportions of HG region in hawthorn pectins, even higher than those reported in this study, which together with the low amount of arabinose and galactose, may evidence that some side chains have been cleaved during their extraction procedure and removed during dialysis. The HG predominance was supported by the HG/RG-I ratio, which followed the order $MI+E > DA+E > CP2 > CP1$. The molar ratio of $(Ara + Gal)/Rha$, indicative of the degree of branching of RG-I segments (Alba et al., 2015; Houben, Jolie, Fraeye, Loey, & Hendrickx, 2011), showed that RG-I domains of CP2 (5.7) and DA+E (3.1) were notably more branched than CP1 (2.7) and MI+E (2.2). This occurrence suggests shorter side chains in RG-I regions of CP1 and MI+E (Alba et al., 2015). Specifically, DA+E possessed more abundant RG-I domains (lower HG/RG-I ratio) with a higher degree of branching compared to MI+E hawthorn pectin, results that were also supported by their linkage analysis (higher proportion of T-Araf and T-Galp).

3.2 FT-IR spectroscopy

FT-IR spectra were used to identify possible structural differences between the citrus and hawthorn samples and to estimate the degree of methyl esterification (DM) of GalA. FT-IR spectrum was recorded over the wavenumber range of 600 to 4000 cm^{-1} (Figure 3). In

the spectrum, the intense broad band between 3200 cm^{-1} and 3600 cm^{-1} , centered at $\sim 3330\text{ cm}^{-1}$, is characteristic of the stretching vibration of O-H groups due to inter- and intra-molecular hydrogen bonding of the GalA backbone (Kpodo et al., 2017). A weaker absorption band in the region of $3000\text{--}2800\text{ cm}^{-1}$ (with peak at 2900 cm^{-1}) was assigned to C-H absorption; including stretching vibrations of CH, CH₂ and methyl (CH₃) group of the methyl esters (Monsoor, Kalapathy, & Proctor, 2001). This band was less intense for purified hawthorn pectins compared to the crude hawthorn pectins. In addition, bands at $1730\text{--}1720\text{ cm}^{-1}$ are derived from C=O stretching vibration of methyl esterified carboxyl groups whereas bands at $1630\text{--}1600\text{ cm}^{-1}$ are derived from carboxylate anion (COO⁻) stretching vibration, and the relative proportion of their areas can be used to estimate the degree of methyl esterification (DM) of the samples (Manrique & Lajolo, 2002). The relative intensity of the 1730 and 1630 cm^{-1} bands of DA and MI pectins was more similar to those of commercial citrus pectins than to DA+E and MI+E. Purified hawthorn pectins DA+E and MI+E showed a significant decrease in the intensity of the 1730 cm^{-1} band and a further increase in the 1630 cm^{-1} band, denoting a decrease in the DM with purification treatment (Figure 3A and 3B). The DM estimated from the area under these bands, calculated as explained in section 2.2.4, showed values of 61-64 % for DA and MI hawthorn pectins, whereas purified pectins (DA+E and MI+E) exhibited a low degree of methyl esterification (35-38 %). The significantly lower DM resulting from the purification step to remove starch molecules could be related to the prolonged times at high temperatures during starch hydrolysis. Levigne, Ralet, and Thibault (2002) found that the highest DM values of sugar beet pectins were obtained for shorter extraction times. Similarly, Pereira et al. (2016) reported that lower DM values (DM = 54 %) resulted from extraction at higher temperatures ($86\text{ }^{\circ}\text{C}$) and times (128 min), which, in turn, enhanced the yield of GalA in pectin from pomegranate peels. Therefore, the longer

times at relatively high temperatures used in our study for pectin purification during starch hydrolysis conditions (240 min and 62.5 °C) may be responsible for the lower DM values. Thus, these results were explained based on a chemical de-methyl esterification due to harsher extraction conditions and even to a possibly better extraction of more strongly bound pectins, possibly of lower DM, which would also be in agreement with the slightly higher protein values found in purified hawthorn pectins. Another explanation could be that during the prolonged time at 62.5 °C, less acidic pH and lower temperatures used for starch removal, endogenous pectin methyl esterase (PME) enzymes in the hawthorn samples could have been reactivated, as high citric acid concentrations used in crude pectin extraction have been reported to inhibit PME activity (Kurita, Fujiwara, & Yamazaki, 2008). Nonetheless, this option should be disregarded as, in this work, all hawthorn pectins were initially incubated at 90 °C for 2 h in citric acid, time and temperature long/high enough to completely inactivate PME (i.e. PME complete inactivates at 80 °C for 10 min) (Vivar-Mera, Salazar-Montoya, Calva-Calva, & Ramos-Ramírez, 2007). Therefore, chemically induced de-methyl esterification (acidic demethoxylation) was suggested as the most plausible mechanism for the reduction in the DM (Einhorn-Stoll, Kastner, Urbisch, Kroh, & Drusch, 2019). In this regard, high- (DM > 70 %) and low-methoxyl (DM < 38 %) pectins have been extracted from hawthorn fruits (Cuevas-Bernardino et al., 2016; Guo et al., 2019; Jiang et al., 2018; Sun, Chen, & Zhu, 2020), confirming that the DM was primarily governed by the extraction protocol (Levigne et al., 2002; Pereira et al., 2016).

The DM of high methoxyl hawthorn and citrus pectins was corroborated with titration measurements, where DA (76.9 ± 0.0 %) was reported to have significantly higher DM than MI (70.3 ± 0.2 %) pectin. For citrus pectins, CP2 was reported to have significantly higher DM than CP1, (79.9 ± 0.1 % and 67.9 ± 0.3 %, respectively), which is in line with

the values reported by the manufacturer, included in materials section. Likewise, titration results confirmed that purification treatment resulted in low methoxyl hawthorn pectins, with DM values of $33.6 \pm 0.2 \%$ and $34.5 \pm 0.1 \%$ for DA+E and MI+E, respectively. Spectral bands due to the amide I and II of proteins typically absorbing at around 1651 cm^{-1} and 1555 cm^{-1} , respectively, were not observed in any of the samples. This is in accordance with the protein content determined in these samples ($< 4.4 \%$, d.b.). In this sense, the relatively low amount of protein compared to pectin, rather than an overlap between their vibration bands, could be the reason why proteins were not detected in the spectra of crude or purified pectins (Kpodo et al., 2017). Nonetheless, the protein content in the pectin samples ($2.9 - 4.4 \%$, d.b.) could be high enough to enhance and facilitate the interfacial adsorption of pectin at the oil-water interface and, therefore, to confer pectic polysaccharides with excellent emulsifying properties. The bands at approximately 1420 , 1380 , 1320 cm^{-1} tentatively arising from the CH, OH or CH_2 bending and 1230 cm^{-1} from stretching of $-\text{CH}_3\text{CO}$ were followed by high absorption bands between 1200 and 900 cm^{-1} , referred to as the fingerprint region, specific for each polysaccharide (Kpodo et al., 2017; Manrique & Lajolo, 2002; Pereira et al., 2016). The fingerprint region can be indicative of variations in neutral sugar composition of pectins due to the presence of vibration bands attributed to monosaccharide components (Kpodo et al., 2017). In the fingerprint region, well defined absorption peaks were observed at ~ 1140 , 1095 , 1072 , 1043 , 1017 and 920 cm^{-1} for pectin samples, especially in CP1 and CP2 pectins and purified hawthorn pectins. Similar absorption bands were found in hawthorn pectins with high GalA content ($> 83 \%$) by Jiang et al. (2018). Bands at ~ 1140 and $\sim 1040 \text{ cm}^{-1}$ were assigned to stretching vibrations of glycosidic bonds (C-O) and pyranoid rings (C-C) (Kamnev, Colina, Rodriguez, Ptitchkina, & Ignatov, 1998). A reduction of the sharp peak at 1017 cm^{-1} and a more defined peak at 1043 cm^{-1} were observed in this region with

purification of hawthorn pectins (DA+E and MI+E), resulting in absorption profiles more alike to CP1 and CP2 pectins. The several absorption peaks between ~ 1017 and 1103 cm^{-1} indicated the presence of pyranoses and furanoses in pectin samples (Jiang et al., 2018), suggesting that DA and MI contained higher amounts of these than DA+E and MI+E counterparts, possibly due to glucose (i.e. starch) elimination during purification process.

3.3 ^1H -NMR spectroscopy

The overlapped ^1H NMR spectra of the hawthorn and citrus pectins are given in Figure 4, where G and E indicate chemical shifts associated with non-esterified and esterified units, respectively. Although hawthorn and citrus pectins have a similar spectra profile, the detailed study shows significant spectral differences due to their different DM. In all of them, a common sharp singlet at 3.78 ppm assigned to the protons of the methyl groups of esterified pectin is observed; however, the intensity is different depending on the methylation degree of the sample. H-2 and H-3 protons of the GalA residues give the broad signals centered at 3.743 ppm and 3.979 ppm respectively, these chemical shifts being poorly affected by the methylation of the carboxylic acid residues (Rosenbohm, Lundt, Christensen, & Young, 2003; Winning, Viereck, Nørgaard, Larsen, & Engelsen, 2007). At the chemical shift of H-3, differences were mainly observed for DA and MI, with the presence of a triplet not observed in the rest of samples, which may be attributed to its lower purity, and, hence, lower yield in GalA. Chemical shifts of the H-1, H-4 and H-5 GalA protons are dependent on their esterification and the nature of the neighbouring units (Andersen, Larsen, & Grasdalen, 1995). Thus, as observed in CP1, CP2, DA and MI samples, a broad signal at 4.453 ppm due to the H-4 protons of the esterified (E) GalA is clearly observed whereas in MI+E and DA+E an additional downshifted signal is visible at 4.395 ppm. This latter signal is assigned to the H-4 protons of the non-esterified

(G) GalA residues of the pectin (Petersen, Meier, Duus, & Clausen, 2008; Renard & Jarvis, 1999). According to Andersen et al. (1995), H-1 protons of two esterified (EE) GalA units give a signal at 4.906 ppm and one esterified unit linked to a non-esterified (EG) at 4.954 ppm. Signals of non-esterified units (GG) and the transition from non-esterified to esterified units (GE) arises at higher chemical shifts (from 5.11 to 5.15 ppm), as shown in Figure 4. Finally, H-5 GalA proton resonance also shows a clear dependence on the esterification state of the pectin, protons of non-methylated (G) units arising at 4.667 ppm (see DA+E and MI+E spectra) and esterified (E) ones between 5.00 and 5.11 ppm (Andersen et al., 1995; Petersen et al, 2008; Renard & Jarvis, 1999). Thus, the detailed observation of the ^1H NMR signals in MI+E and DA+E spectra shows a similar methylation degree of these two purified pectins and a significant lower level in comparison with the known high methoxyl CP1 and CP2 samples and the DA and MI counterparts. In addition, results for DM obtained by ^1H NMR are in total agreement with those of FTIR analysis shown in Table 1.

The GalA of the backbone can be also O-acetylated at C-2 or C-3. The signal in the range of 2.05-2.13 ppm (supplementary material 1), which is indicative of O-acetyl substituents (Alba et al., 2015), was low for all the samples, compared to acetylated okra pectin, also included for comparison. Thus, the presence of acetylation in hawthorn and citrus pectins was not significant. The degree of acetylation for citrus pectin has been reported to be low (1.4-1.6 %) compared to other sources such as beet (Leroux et al., 2003) or okra (Kpodo et al., 2017) pectins, with reported degree of acetylation > 16 % and > 20 %, respectively. Similarly, Linares-Garcia, Ramos-Ramirez and Salazar-Montoya (2015) found that the degree of acetyl-esterification for hawthorn and citrus pectins was 1.9 and 2.3 %, respectively, which would agree with the negligible signals found in the spectra.

3.4 Molecular weight determination

Molecular weight of the isolated hawthorn and commercial citrus pectins was evaluated by size exclusion chromatography, with elution profiles of refractive index and average molecular weight (M_w) shown in supplementary material 2 and Table 1, respectively.

For high DM pectins, M_w of CP2 (235 kDa) was higher than CP1 (164 kDa), whereas for crude hawthorn pectins, the M_w of MI (213 kDa) was higher than that from DA (159 kDa). The M_w of citrus pectin was in agreement with those values reported in literature (162-217 kDa) (Leroux et al., 2003; Morris, Foster, & Harding, 2000). In the case of hawthorn berries, pectic polysaccharides with weight average molecular weight between 111.7 and 295.2 kDa have been previously reported depending on the extraction procedure (Cuevas-Bernardino et al., 2016; Guo et al., 2019; Jiang et al., 2018).

For purified hawthorn pectins, both DA+E (126 kDa) and MI+E (98 kDa) presented lower M_w than their corresponding high DM hawthorn counterparts. A shift towards higher retention times was observed for the first broad peak in the elution profiles of purified hawthorn pectins together with a more prominent second smaller peak at higher retention times (see supplementary material 2). This variation in elution patterns with a narrower first peak appearing at higher retention times, and resulting in a lower M_w of purified pectins, could be attributed to either or both the removal of co-eluting starch and the partial hydrolysis of hawthorn pectins. As for the first hypothesis, starch is supposed to have larger size than pectin (Bertoft, 2017), which could be co-eluting with some pectin aggregates in crude pectin, contributing to a first broader peak in DA and MI. Therefore, M_w for crude hawthorn pectins measured before starch purification cannot accurately reflect the M_w of hawthorn pectins. However, a second hypothesis supporting a chemical degradation of hawthorn pectins during the additional step for starch removal (i.e., high temperature, low pH and long extraction times) is also plausible. In fact, the chemical demethoxylation of pectin, which occurred during the enzymatic step of starch removal

(Table 1), could be taking place in parallel with both/either acid hydrolysis (i.e., acid-catalyzed cleavage of RG-I and/or HG backbone) and/or β -elimination of the HG backbone, resulting in pectin with lower DM and M_w (Diaz, Anthon, & Barrett, 2007; Einhorn-Stoll et al., 2019; Kurita et al., 2008). The extent of each of these two events that lead to chemical degradation is dependent on the initial DM and extraction conditions (Diaz et al., 2007; Einhorn-Stoll et al., 2019). Specifically, at pH about 4.5 and $T \geq 60^\circ\text{C}$, acid hydrolysis occurs more rapidly than β -elimination, the later preferentially occurring at higher temperatures and more basic pH (Diaz et al., 2007; Ngouémazong, Christiaens, Shpigelman, Van Loey, & Hendrickx, 2015). Moreover, β -elimination rate also decreases as DM decreases (Chan et al., 2017; Diaz et al., 2007). Thus, the long extraction times at relatively low temperature and at $\text{pH} < 4.5$, attained during starch removal, should favor the occurrence of acid hydrolysis over β -elimination. Although pectin de-methoxylation indirectly enhances its depolymerization reaction, the reaction rate of acidic demethoxylation is higher than that of acidic hydrolysis, especially of the HG backbone (Einhorn-Stoll et al., 2019). Therefore, purification of hawthorn pectins under those conditions might result in chemical de-methoxylation without a drastic de-polymerization of the pectin backbone (Chan et al., 2017), as suggested by the high GalA content of the samples (Table 1). A schematic representation of the previous discussions is depicted in Figure 5.

3.5 Emulsifying properties

The emulsifying potential of hawthorn and citrus pectins were expressed as emulsifying activity (EA) and emulsion stability (ES) and the results are shown in Figure 6. Broadly speaking, the emulsifying functionality of pectin molecules is controlled by the structural characteristics of the polysaccharide, including features such as the molecular weight, methyl, feruloyl and acetyl content, proportion of HG domain, conformation of RG-I side

chains, and presence of attached protein groups (Kpodo et al., 2018; Ngouémazong et al., 2015). In oil in water emulsions, the hydrophobic protein material adsorbs at the oil-water interface, serving as anchoring point, with only a small fraction of the pectin molecules being adsorbed at the interface (Cuevas-Bernardino et al., 2016). The oil droplets are also sterically stabilized by the hydrophilic carbohydrate moieties protruding into the aqueous phase (Gülseren & Corredig, 2014). The remaining pectin molecules will also contribute to the decrease in the tendency of the dispersed oil droplets to approach each other by increasing the viscosity of the continuous phase (Cuevas-Bernardino et al., 2016).

Among all pectin samples, purified DA+E and MI+E hawthorn pectins resulted in greater EA than crude hawthorn pectins and commercially purified pectins from citrus peel. It is well known that dispersed starch molecules are highly hydrophilic and, therefore, do not tend to concentrate at the interface of oil-in-water (BeMiller, 2018). In other words, they lack of surfactant properties. On the other hand, pectic polysaccharides have been reported to form and stabilize emulsions (Alba et al., 2013; Cuevas-Bernardino et al., 2016; Schmidt et al., 2015, Xu et al., 2020). Therefore, removing starch and increasing the concentration of pectin extracted is paramount to enhance the surfactant potential of the pectic material. In this study, when DA and MI were removed for correlation analyses due to high content of starch impurities (that also possesses thickening ability), EA of purified citrus and hawthorn pectins (GalA > 77 %) correlated positively with the proportion of HG ($r = 0.983$, $p < 0.05$) and negatively with the proportion of RG-I ($r = -0.986$, $p < 0.05$) of the pectin structure. The RG-I region is ascribed to provide steric stabilization, whereas the HG domain is associated with electrostatic stabilization (Ngouémazong et al., 2015). According to Ralet et al. (2008), chain flexibility has been found to be higher for pectin chains with greater content in RG-I regions. This additional flexibility of the chains would create shorter separation distances between droplets

resulting in ineffective steric stabilization and faster pectin desorption compared to those with lower amounts of RG-I (Kpodo et al., 2008). Thus, the proportion of RG-I found in hawthorn and citrus samples (Table 3) could be the reason for the differences found in EA. In addition, the mutual repulsion between the ionized carboxylic groups in HG may cause repulsion between oil droplets avoiding fast coalescence during emulsion formation (Ngouémazong et al., 2015).

For the stability of the formed emulsions, samples were destabilized at adverse conditions of high temperature to accelerate the coalescence and further creaming phenomena in the emulsions. After heating the emulsions, crude and purified hawthorn pectins also presented generally higher ES than citrus pectins. When DA and MI were removed from the correlations analysis, a higher ES was observed for pectin samples with higher amount of protein ($r=0.977$ and $p<0.05$). The hydrophobic covalently bound proteins of pectin adsorb at the oil-water interface and has been reported to be a major determinant for the emulsifying potential of pectin (Schmidt et al., 2015). Thus, bonded-protein could enhance and facilitate the interfacial adsorption of large macromolecules at the oil-water interface, providing better stability. In this regard, fast adsorption to the interface has been described as a critical step for emulsion stabilization (Schmidt et al., 2015). Perhaps, a higher amount of proteinaceous material could also delay pectin desorption from the oil-water interface, making emulsions more stable. In this sense, some authors have reported that beet and citrus pectins were capable of stabilizing oil-in-water emulsions, with the small amount of covalently-bound protein playing a key role due to their adsorption at the oil-water interface and the creation and stabilization of smaller droplets (Leroux et al., 2003). It is noteworthy that the protein content in CP1 and MI+E was not statistically different ($p > 0.05$) and, therefore, ES cannot be solely explained on the protein content. Thus, results suggest that it is the combination of a higher protein content (delaying pectin

desorption from the interface) and a higher proportion of HG domains with lower DM (conferring electrostatic stabilization) that might jointly improve the emulsifying properties of hawthorn pectin materials.

4 Conclusions

In this work, two Chinese Hawthorn (*Crataegus pinnatifida*) cultivars were investigated, namely one of the most common cultivars (Dajinxing, DA) and a less common cultivar that is gaining popularity for its larger and softer fruits (Mianqiu, MI). DA and MI yielded pectin with a significantly higher proportion of the homogalacturonan domain and a significantly better potential to form and stabilize emulsions than purified pectins from citrus peel. However, MI berry presented a significantly lower starch content and, as a consequence, it resulted in a 2-fold greater yield of pectin than DA cultivar. This occurrence is critical for successful utilization of hawthorn berries and their by-products from the wine and juice industry into raw materials for manufacturing of purified pectin. Remarkably, this work also shows that purified hawthorn pectins (GalA content > 83%) can be obtained through hot acid extraction followed by an enzymatic step for starch removal, which, in turn, results in pectin demethoxylation (most likely acidic demethoxylation). Hence, small variations in the complex molecular structure of hawthorn pectins occurring during extraction and purification could significantly influence their functional properties, such as their gelling properties (high vs low methoxyl gelation behavior) and suitability as emulsifiers.

Although DA and MI purified pectins did not exhibit differences in their emulsifying performance, further colloidal studies are needed to exploit the structural differences of DA and MI pectins. Remarkably, results showed that MI presents a higher proportion of the homogalacturonan domain, less branched rhamnogalacturonan I domains, and

significantly shorter arabinan side chains than pectin from DA, which might result in critical differences in functionality as food ingredients and biomaterials.

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Credit authorship contribution statement

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodhyd.2020.106476>.

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Tables

Table 1. Galacturonic acid and protein contents, degree of methyl esterification and molecular weight of citrus and hawthorn pectic polysaccharides.

	Galacturonic Acid (% weight, db)	Protein (%, db)	DM (FTIR) (%)	DM (titration) (%)	M _w (kDa)
CP1	77.4 ±1.1c	3.8 ±0.1c	75 ±1c	67.9 ±0.3b	164.0 ±2.8c
CP2	86.4 ±1.4d	2.8 ±0.0a	75 ±1c	79.9 ±0.1e	234.5 ±10.1d
DA	38.3 ±2.5a	3.1 ±0.0b	64 ±0b	76.9 ±0.0d	159.0 ±2.8c
MI	61.2 ±2.8b	2.9 ±0.1a	61 ±0b	70.3 ±0.2c	213.0 ±7.1e
DA+E	83.2 ±0.5d	4.4 ±0.0d	35 ±0a	33.6±0.2a	126.0 ±5.6b
MI+E	85.3 ±1.4d	3.9 ±0.0c	38 ±0a	34.5±0.1a	97.5 ±0.2a

Values with different letters in the same column are significantly different with $p < 0.05$. Degree of methyl esterification (DM, %) determined with FTIR and titration methods, respectively. Molecular weight (M_w) determined by size exclusion chromatography. CP1, commercial high methoxyl citrus pectin 1; CP2, commercial high methoxyl citrus pectin 2; DA, crude Dajinxing hawthorn pectin; MI, crude Mianqiu hawthorn pectin; DA+E, purified Dajinxing hawthorn pectin with enzymatic treatment; MI+E, purified Mianqiu hawthorn pectin with enzymatic treatment.

Table 2. Neutral sugar and glycosyl-linkage composition (mol %) of crude and purified hawthorn pectins.

	Monosaccharide (mol %)				Glycosyl Linkage	Linkage (mol %)			
	DA	MI	DA + E	MI + E		DA	MI	DA + E	MI + E
Rhamnose	1.09 ±0.12a	1.31 ±0.14a	1.95 ±0.05b	2.06 ±0.05b	T-Rhap	0.03 (3)	0.04 (3)	0.05 (3)	0.05 (3)
					1,2-Rhap	0.68 (62)	0.67 (51)	1.04 (53)	1.28 (62)
					1,2,4-Rhap	0.38 (35)	0.61 (47)	0.86 (44)	0.73 (35)
Fucose	0.35 ±0.02a	0.36 ±0.09a	0.34 ±0.02a	0.30 ±0.01a	T-Fucp	0.35 (100)	0.36 (100)	0.34 (100)	0.30 (100)
Arabinose	1.69 ±0.09a	1.68 ±0.00a	2.13 ±0.05a	2.35 ±0.36a	T-Araf	0.56 (33)	0.55 (33)	0.50 (24)	0.85 (36)
					T-Arap	0.10 (6)	0.10 (6)	0.08 (4)	0.06 (2)
					1,3-Araf	0.12 (7)	0.08 (5)	0.17 (8)	0.10 (4)
					1,5-Araf	0.68 (40)	0.44 (26)	0.94 (44)	1.01 (43)
					1,2-Arap	0.24 (14)	0.50 (30)	0.43 (20)	0.33 (14)
Xylose	0.77 ±0.04a	0.94 ±0.07ab	1.15 ±0.07b	1.15 ±0.08b	T-Xylp	0.77 (100)	0.94 (100)	1.15 (100)	1.15 (100)
Mannose	0.35 ±0.04a	0.38 ±0.08a	0.78 ±0.02c	0.58 ±0.06b	T-Manp	0.35 (100)	0.38 (100)	0.78 (100)	0.58 (100)
Galactose	1.94 ±0.13a	3.23 ±0.08b	3.99 ±0.22b	2.24 ±0.59a	T-Galp	0.35 (18)	0.49 (15)	0.65 (16)	0.48 (21)
					1,2-Galp	0.04 (2)	0.09 (3)	0.10 (2)	0.05 (2)
					1,3-Galp	0.34 (18)	0.37 (11)	0.28 (7)	0.27 (12)
					1,4-Galp	1.02 (52)	2.11 (65)	2.66 (67)	1.19 (53)
					1,6-Galp	0.19 (10)	0.18 (6)	0.30 (8)	0.24 (11)
Glucose	58.90 ±3.84c	35.27 ±3.64b	1.43 ±0.26a	1.28 ±0.25a	T-Glcp	4.18 (7)	2.30 (7)	0.11 (8)	0.14 (11)
					1,4-Glcp	49.20 (84)	29.69 (84)	0.79 (55)	0.84 (66)
					1,2,4-Glcp	0.31 (1)	0.21 (1)	0.03 (2)	0.01 (1)
					1,3,4-Glcp	1.17 (2)	0.65 (2)	0.05 (4)	0.09 (7)
					1,4,6-Glcp	4.04 (7)	2.42 (7)	0.45 (31)	0.20 (16)

Values with different letters in the same row are significantly different with $p < 0.05$. Number in parentheses are linkage percentage relative to 100% of each monomer. *f* and *p* for each glycosyl linkage indicate the presence of the neutral sugar as furanose and pyranose forms, respectively. DA, crude Dajinxing hawthorn pectin; MI, crude Mianqiu hawthorn pectin; DA+E, purified Dajinxing hawthorn pectin with enzymatic treatment; MI+E, purified Mianqiu hawthorn pectin with enzymatic treatment.

Table 3. Percentage of homogalacturonan (HG) and rhamnogalacturonan-I (RG-I) on molar basis of pectin powder, relative molar ratio of HG and RG-I regions (HG/RG-I) and degree of branching (DB) of RG-I for citrus and purified hawthorn pectins.

	HG	RG-I	HG/RG-I	DB of RG-I
CP1	79.0	18.7	4.2	2.7
CP2	83.0	14.7	5.6	5.7
DA+E	86.3	10.0	8.6	3.1
MI+E	88.0	8.7	10.1	2.2

$$\text{HG (mol\%)} = \text{GalA (mol \%)} - \text{Rha (mol \%)}$$

$$\text{RG-I (mol \%)} = 2 * \text{Rha (mol \%)} + \text{Ara (mol \%)} + \text{Gal (mol \%)}$$

$$\text{DB of RG-I (mol \%)} = [\text{Ara (mol \%)} + \text{Gal (mol \%)}] / \text{Rha (mol \%)}$$

CP1, commercial high methoxyl citrus pectin 1; CP2, commercial high methoxyl citrus pectin 2; DA+E, purified Dajinxing hawthorn pectin with enzymatic treatment; MI+E, purified Mianqiu hawthorn pectin with enzymatic treatment.

Figures

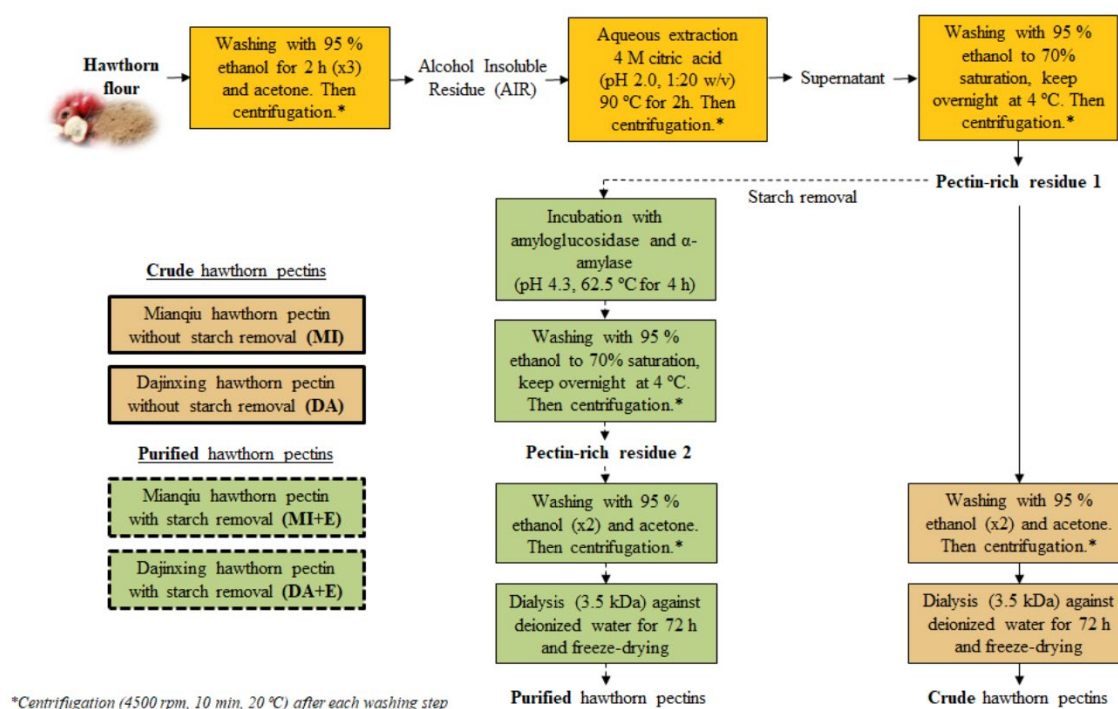


Fig. 1. Flow chart depicting the isolation procedure followed for crude (DA and MI) and purified (DA+E and MI+E) hawthorn pectins from Dajinxing (DA) and Mianqiu (MI) cultivars.

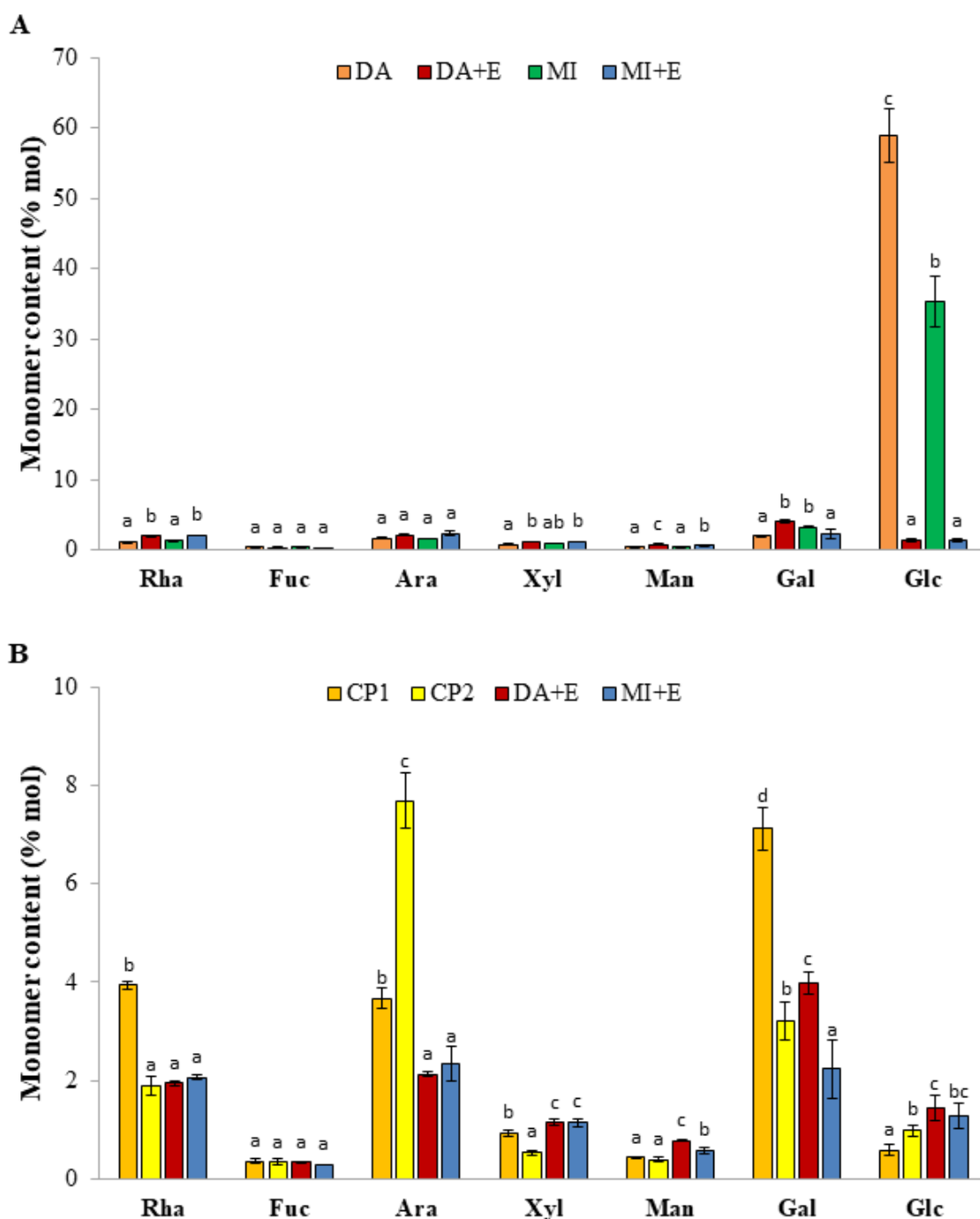


Fig. 2. Neutral sugar composition as molar percentage of total monomers content. Monomer composition of hawthorn obtained from Dajinxing (DA) and Mianqiu (MI) cultivars (A), before and after starch removal with enzymatic treatment (+ E). Monomer composition of purified hawthorn pectins (+ E) and citrus pectins (B). CP1, commercial high methoxyl citrus pectin 1; CP2, commercial high methoxyl citrus pectin 2; DA, crude Dajinxing hawthorn pectin; MI, crude Mianqiu hawthorn pectin; DA+E, purified Dajinxing hawthorn pectin with enzymatic treatment; MI+E, purified Mianqiu hawthorn pectin with enzymatic treatment.

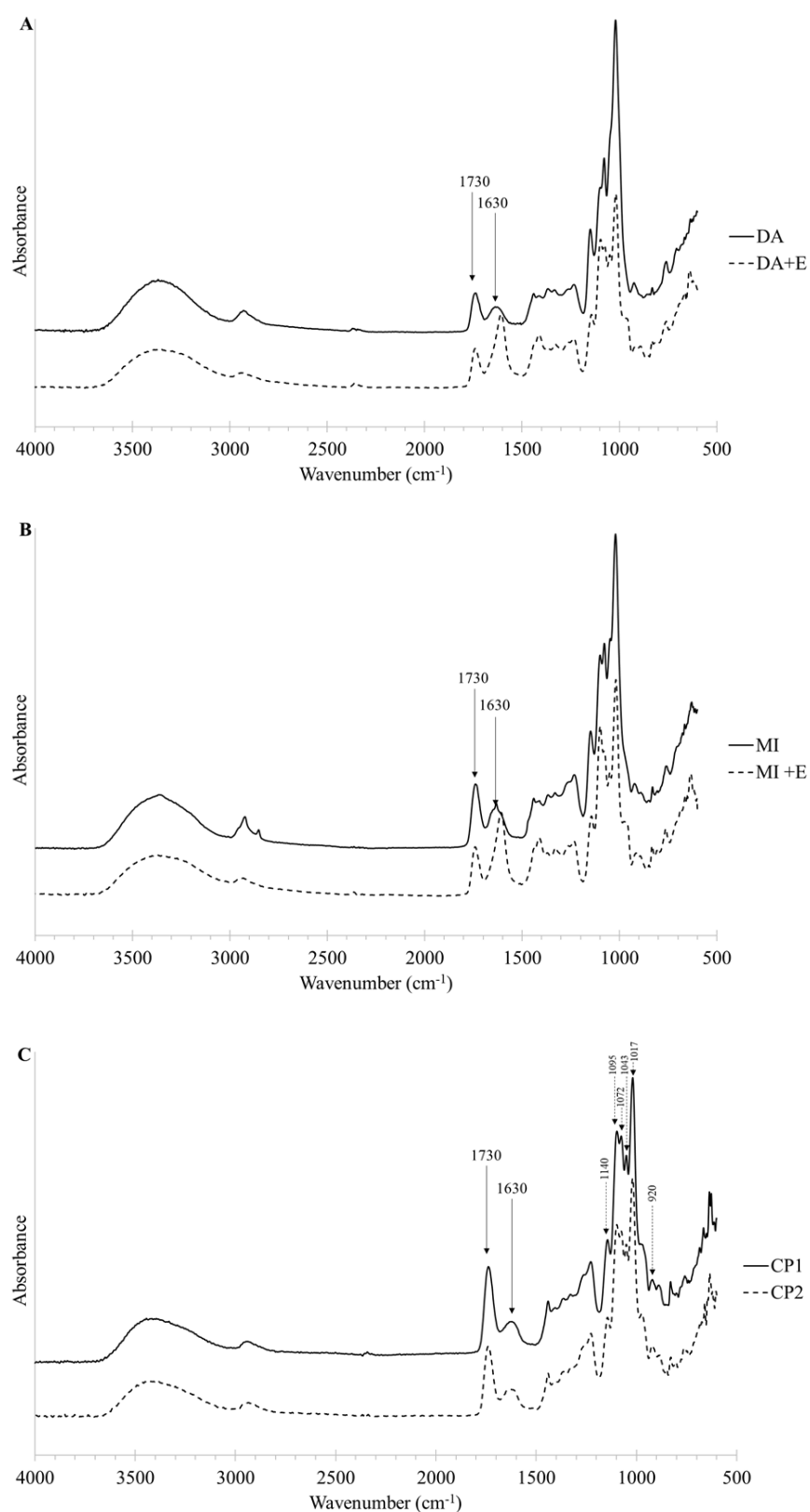


Fig. 3. FTIR spectra of hawthorn (A and B) and citrus (C) pectins. 1730 and 1630 arrows indicate the absorption bands used for calculation of degree of methyl esterification. Arrows with discontinuous lines in Figure C highlight well defined absorption peaks in the fingerprint region of pectins. CP1, commercial high methoxyl citrus pectin 1; CP2,

commercial high methoxyl citrus pectin 2; DA, crude Dajinxing hawthorn pectin; MI, crude Mianqiu hawthorn pectin; DA+E, purified Dajinxing hawthorn pectin with enzymatic treatment; MI+E, purified Mianqiu hawthorn pectin with enzymatic treatment.

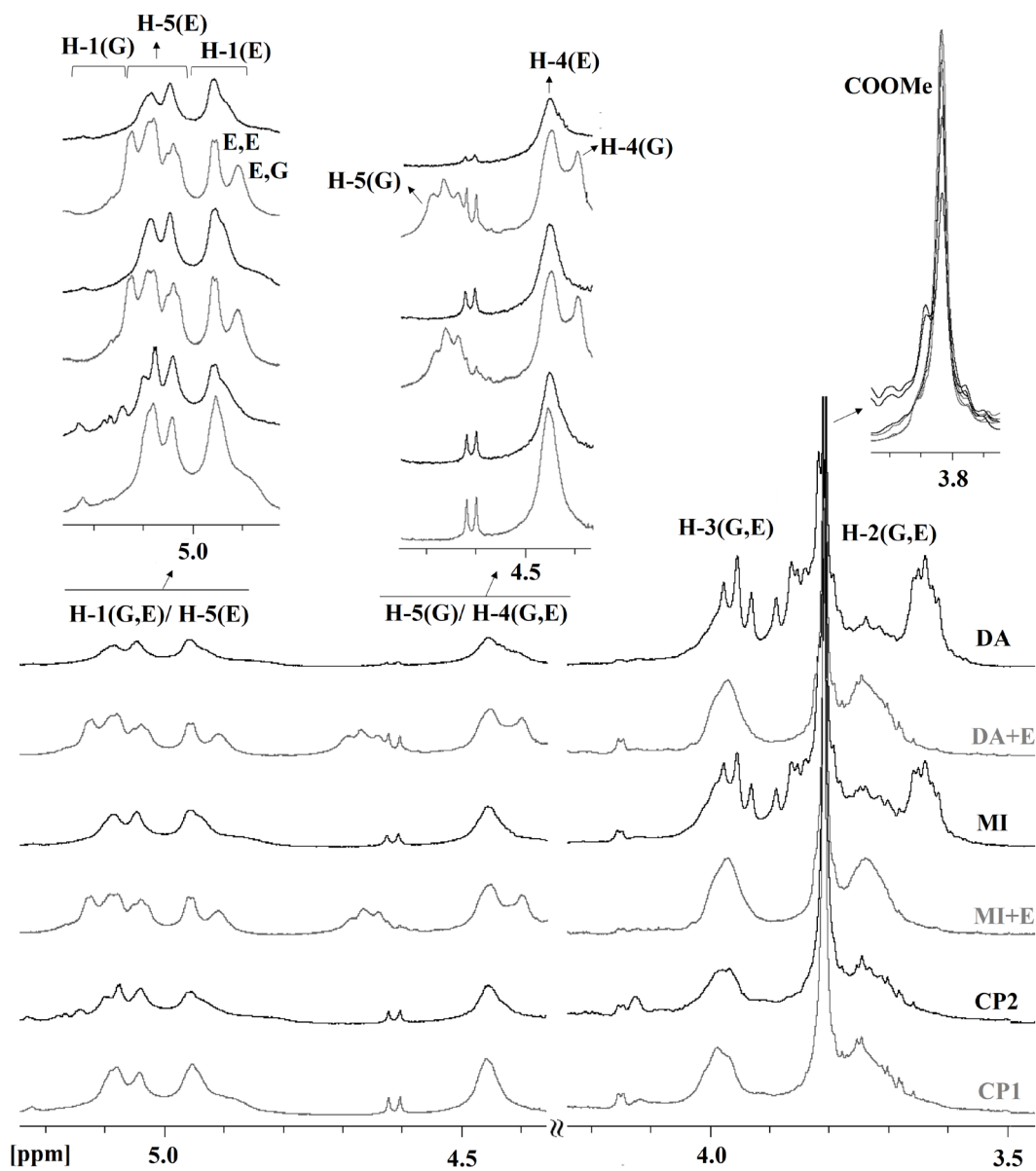


Fig. 4. ^1H NMR spectra region between 3.5 and 5.2 ppm of crude (DA and MI) and purified (DA+E and MI+E) pectins from Dajinxing (DA) and Mianqiu (MI) hawthorn flours and commercial citrus pectins (CP1 and CP2) with different methyl esterification degree, together with the enlargement of several signals. G and E represent non-esterified and esterified units, respectively.

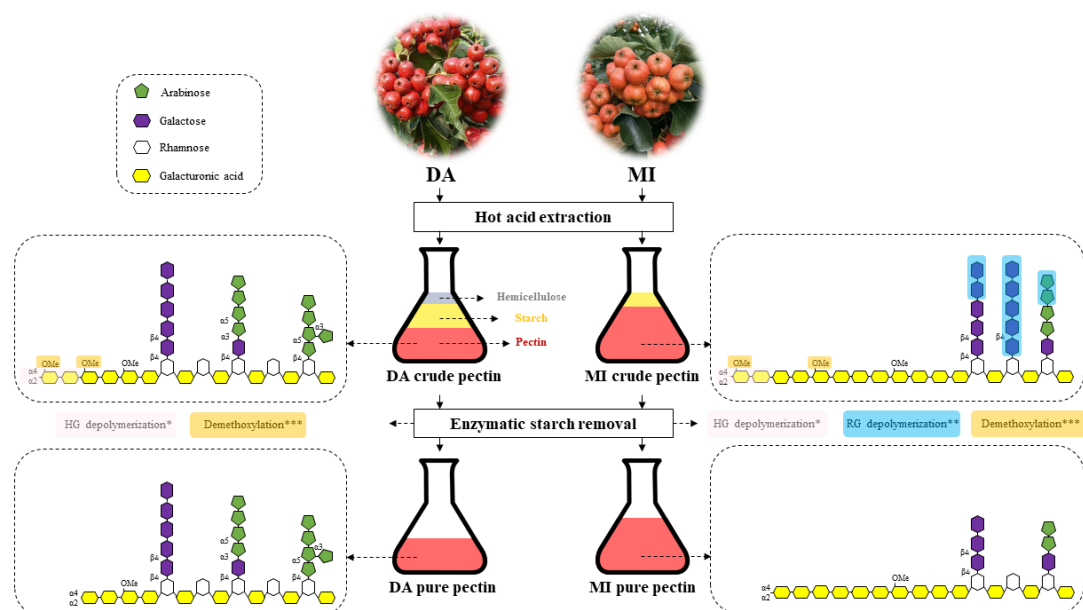


Fig. 5. Schematic representation of the yield and molecular structures of the different crude and purified pectins extracted from Dajinxing (DA) and Mianqiu (MI) cultivars of Hawthorn berries. HG, Homogalacturonan; RG-I, Rhamnogalacturonan I. Asterisks represent the degree of evidence for the proposed phenomena occurring during the purification step, namely 1, 2 and 3 for weak, moderate and strong evidence, respectively. Only the most abundant glycan monomers are represented for easier visualization of major structural differences.

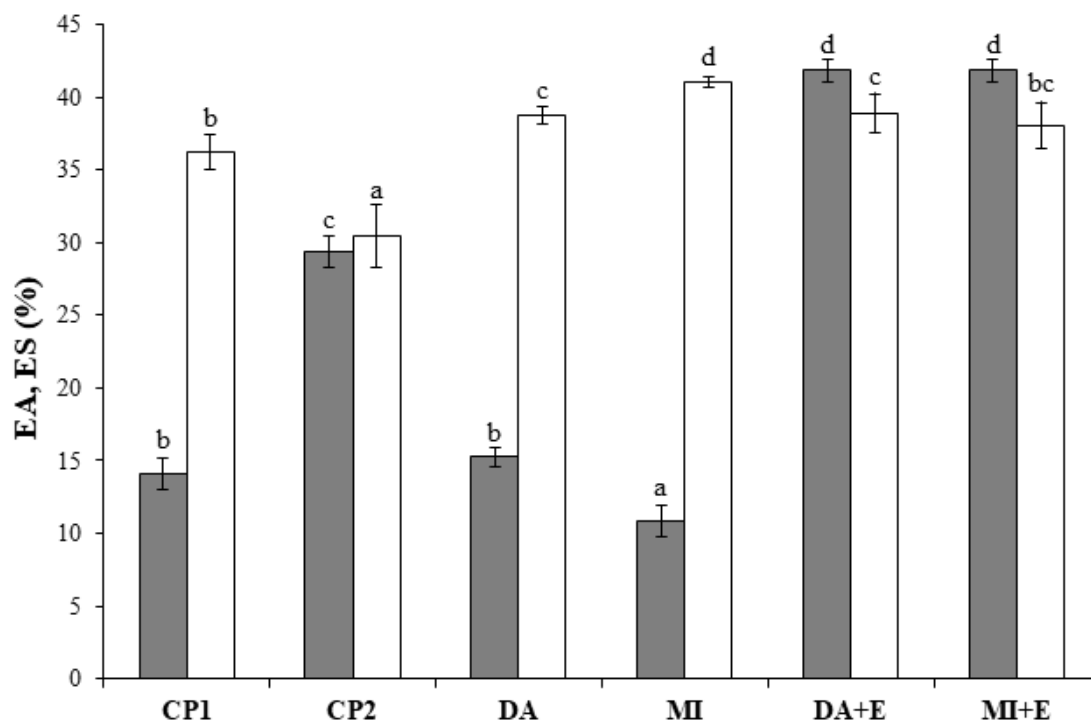


Fig. 6. Emulsifying ability (EA) and emulsion stability (ES) of citrus and hawthorn pectins in grey and white bars, respectively. CP1, commercial high methoxyl citrus pectin 1; CP2, commercial high methoxyl citrus pectin 2; DA, crude Dajinxing hawthorn pectin; MI, crude Mianqiu hawthorn pectin; DA+E, purified Dajinxing hawthorn pectin with enzymatic treatment; MI+E, purified Mianqiu hawthorn pectin with enzymatic treatment.